

Adolfo De Unánue T. | Curriculum Vitæ

🏠 Mexico City, Mexico • 📞 +52-55-2128-7388 • 📧 nanounanue • 📧 nanounanue • 📧 nanounanue • ✉️ adolfo@unanue.mx

ABOUT ME

Senior AI/ML Leader | 15+ years taking programs from research to production across 6 countries

- **Strategy & org building:** Founded and led the Data Science Center (ITESM); created Mexico's first MSc in Data Science (ITAM).
- **Delivery at scale:** Production ML systems for government, healthcare, and justice—interpretable models, fairness audits, governance, and monitoring.
- **Team leadership:** Built and managed cross-functional DS/ML/DE teams across MX/US; P&L and technical direction ownership.
- **Founder/operator:** Co-founded 3 AI ventures (Dialetheia, Open Intelligence, Sm4rt); simulation-driven decisions (SD/ABM/DES) and tech transfer to government and enterprise partners.

LEADERSHIP SNAPSHOT

- **Team:** 18 FTE (DS/ML/DE), cross-functional MX/US **Budget:** Full P&L ownership **Production:** ~35 models/services, 99.9% avail., p95 ≤ 250 ms
- **Reach:** 20+ projects across 6 countries for government agencies, C-suites, and international organizations **Governance:** Aequitas audits, MRM reviews, responsible AI
- **Policy:** Invited expert, Mexican Congress & Senate on AI regulation (2025).

EXPERIENCE

EGOBIERNOYTP @ ITESM | AI Research Director, former Associate Dean 2020 - Present | Mexico City, México

- Associate Dean → AI Research Director. Founded the *Centro de Ciencia de Datos e IA* initiative; stood up the Data Science Center delivering ML from pilots to production (governance, explainability, monitoring).
- Portfolio: case-management ML & DV risk prevention (Zacatecas DA Office), customer process optimization & CV (health), pharma logistics optimization, port/terminal optimization, school siting (El Salvador 2023), 5G nationwide planning (El Salvador).

INDEPENDENT CONSULTING | CTO-as-a-Service, ML Advisory, Legal-Tech 2025 - Present | Mexico City, México

- Legal-ops automation with LLMs & agentic workflows; fractional CTO for AI startups; ML advisory for early-stage ventures.

DSAPP / DSSG | UChicago (2016–2019) & CMU / Alan Turing Inst. (2020–2022) Lead/Senior ML Researcher & Eng. Manager 2016 - 2022 | U.S. / U.K. / LATAM

- Led or contributed to 20 data-science projects across 6 countries: EMS optimization, school dropout prevention (El Salvador 2018), diabetes screening, HIV care retention, homelessness prevention, housing safety, inspections prioritization, police EWS, suicide-lifeline routing, pollution enforcement (NY), public-services matching, urban mobility (Seattle), transport equity (West Midlands, UK), vaccination outreach, and more.
- Co-created *Aequitas* (fairness auditing toolkit) and *Triage* (ML model dev/ops framework)—both adopted across government and nonprofit programs.

ITAM | Director of the CCD / Director MSc Program in DS 2019 - 2020 | Mexico City

- Founded Mexico's first MSc in Data Science and the Centro de Ciencia de Datos (CCD), guiding public and private organizations in decision-science adoption.

OPEN INTELLIGENCE & DATANK & SM4RT PREDICTIVE | Co-Founder/CTO, Chief of Machine Intelligence, Lead DS 2011 - 2017 | Mexico City

- Co-founded/CTO of 3 AI ventures. Built production data products (Arquímedes, Pitágoras) adopted by agencies and consultancies; stood up bank's first fraud DWH—cut detection from weeks to < 1 week, simplified rules by ~90%.
- Enterprise data unification and probabilistic models for C-level decision-making; tech transfer to government and enterprise partners.

TIMELINE

2026 — Health, pharma logistic optimization & DV risk projects (ITESM).
2025 — Invited expert, Congress & Senate on AI regulation.
2025 — Independent consulting: legal-ops, Fractional CTO, ML advisory.
2024 — Case management ML optimization DA Office.
2024 — Terminal/port ML optimization.
2023 — Founded Centro de Ciencia de Datos e IA (ITESM).
2023 — School siting & 5G network design (El Salvador, ITESM).
2022 — Lead DS at DSSG (CMU) — suicide-lifeline, homelessness, roof-damage.
2021 — Lead DS at DSSG (CMU) — vaccination outreach.
2021 — Associate Dean & Research Director EGobiernoYTP (ITESM).
2020 — Founded Centro de Ciencia de Datos (ITAM).

EDUCATION

2010 – 2012 **Postdoctoral Researcher**, Complex Adaptive Systems / DS
 Centro de Ciencias de la Complejidad, UNAM
2006 – 2009 **Ph. D.**, Theoretical Physics
 Universidad Nacional Autónoma de México
2003 – 2006 **M. in Sc.**, Theoretical Physics
 Universidad Nacional Autónoma de México
1998 – 2001 **M. in Sc.**, Electrical Eng. – Mechatronics
 CINVESTAV – IPN
1995 – 1998 **B. Eng.**, Electronics and Control Theory
 Universidad Regiomontana

SELECTED IMPACT

- **Logistics & infrastructure:** Terminal/port ML optimization (Mexico, ITESM); pharma logistics optimization (ITESM); EMS resource placement (Cincinnati, UChicago); school location optimization (El Salvador 2023, ITESM); 5G nationwide design (El Salvador, ITESM); urban mobility (Seattle, UChicago).
- **Health & social:** Proactive diabetes screening (Chicago, UChicago); HIV care retention (Chicago, UChicago); vaccination outreach (Mexico, CMU); social services targeting for SEDESOL (Mexico, UChicago); public-services matching tool (multi-county USA, UChicago).
- **Education:** School dropout prediction & social program optimization (El Salvador 2018, UChicago/DSSG); citizen-request routing (Mexico, UChicago).
- **Housing & homelessness:** Rental-assistance prioritization (Allegheny County, CMU); rough-sleeper service matching (U.K., Alan Turing); roof-damage remediation targeting (Baltimore, CMU).
- **Public safety & justice:** Police EWS/EIS (Charlotte, SF, Nashville, UChicago); case triage with explainability (Zacatecas DA, ITESM); DV risk prevention (Zacatecas, ITESM); legal-ops automation with LLMs (independent); suicide-lifeline routing (CMU).
- **Environment & labor:** Pollution enforcement (NY State, UChicago); workplace safety inspections (Chile, UChicago); transport equity (West Midlands UK, Alan Turing); fostering agency inspections (UK, Alan Turing).

2019 — Lead DS at DSSG (Alan Turing) — transport equity, foster agencies, UK.
2018 — School dropout & diabetes screening (UChicago).
2018 — Workplace safety inspections (Chile, UChicago).
2018 — Urban mobility optimization (Seattle, UChicago).
2017 — Senior Researcher DSaPP; co-created Aequitas & Triage.
2017 — Police EWS (Charlotte, SF, Nashville, Pittsburgh, UChicago).
2016 — Data Science Mentor DSSG (UChicago).
2014 — Endeavor Global Entrepreneur.
2013 — Cofounder Open Intelligence.
2011 — Fraud detection & DWH (Sm4rt / BBVA Bancomer).
2009 — Cofounder Dialetheia Consultores.